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Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gas' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ $= 0,08205 \text{ atm} \cdot \text{L} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero degrees Celsius	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs Energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First-order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Drop in vapor pressure	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of a sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of a sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of a circle	$A_{\text{sp}} = \pi \cdot r^2$	Speed of light	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Long. of a circunf.	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity on Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ N} = \text{kg} \cdot \text{m} \cdot \text{s}^{-2}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$			
$1 \text{ bar} = 100 \text{ kPa}$			
$1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$			
$1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$			
$1 \text{ dm}^3 = 1 \text{ L}$			
$1 \text{ mL} = 1 \text{ cm}^3$			
$1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

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Day 1. Grade 10th. Questions 1, 4, 5 and 6.



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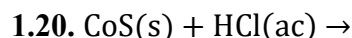
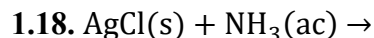
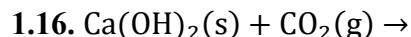
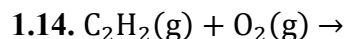
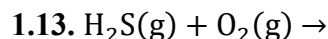
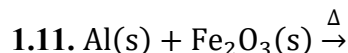
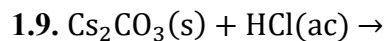
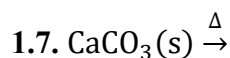
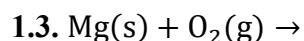
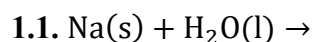
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Question 1. Inorganic reactions (15% of total; 10th grade) Day 1

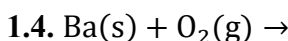
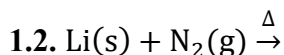
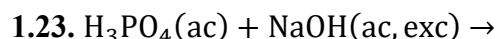
1.1	...	Total	Total
2	Idem	30	15

Chemistry is the science that studies matter, its composition, and its transformations. This discipline did not emerge all at once; it gradually appeared, closely linked to people's daily lives. Since ancient times, people practiced chemistry: although they did not talk about "atoms" or "reactions",

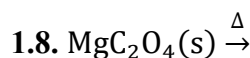
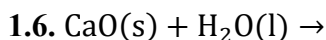
they did know how to transform materials. For example: they used the combustion of wood or coal to cook food, heat themselves and in metallurgy; they obtained bread, wine and beer through fermentation; they mixed metals at high temperatures to create alloys such as bronze; they used dyes and pigments to color fabrics and obtain cosmetics; and they used plants, oils and minerals such as perfumes and medicines. Nowadays, human knowledge has advanced a lot, and it is possible to formulate substances and their transformations. This is one of the most important skills in preparing for the Olympiads. Demonstrate what you have learned in your studies by completing and balancing the following equations.



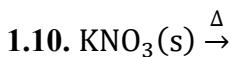
Hint: Products are a gas, a green oxysalt, and a black oxide.



Hint: The oxidation state of oxygen in the product is 1-.



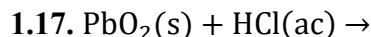
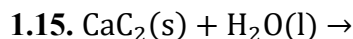
Hint: Two gaseous products are obtained.



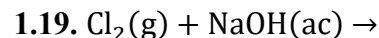
Hint: A new oxysalt is formed.



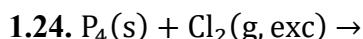
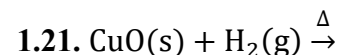
Hint: A brown (reddish-brown) gas is obtained.



Hint: A green-yellow gas with a characteristic odor is obtained.



Hint: A disproportion reaction occurs.



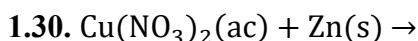
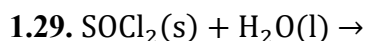
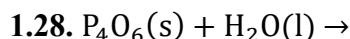
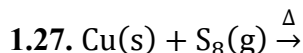
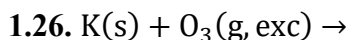
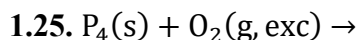
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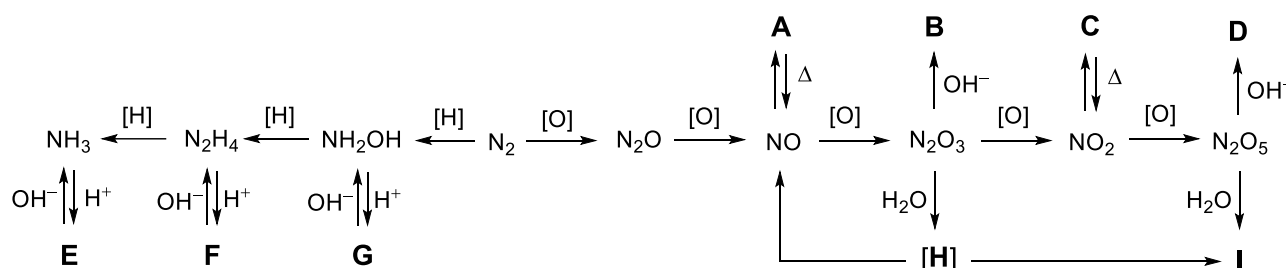
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Question 4. Nitrogen (15% of total; only 10th grade) Day 1

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	Total	Total
18	2	2	2	18	4	6	8	60	15

Nitrogen is an element with a very diverse chemistry because it exhibits oxidation states from -3 to $+5$. The diagram below contains some of the most common chemical species.



- 4.1. Represent the Lewis structures of the nitrogen molecules whose formulas appear in the diagram above.
- 4.2. Why can ammonia, hydrazine, and hydroxylamine react with H^+ ions to form **E**, **F**, and **G** cations?
- 4.3. Why can trioxide and pentoxide molecules react with OH^- ions to form **B** and **D** anions?
- 4.4. Why can monoxide and dioxide molecules form **A** and **C**?
- 4.5. Write the formulas of the substances represented by letters **A-I**.
- 4.6. In concentrated sulfuric medium, the molecules of **H** and **I** dehydrate and form the cations **J** and **K**, respectively, which are highly reactive species. Write down their formulas.
- 4.7. Explain in detail the different behavior of **B** and **D** anions when interacting with transition metals. **B** easily forms complex ions, preferring to bond at N or O when the oxidation state of the metal is low or high, respectively. In contrast, few complex ions of **D** are known.
- 4.8. Construct the diagram of molecular orbitals for NO, assuming that nitrogen has sp hybridization. Identify frontier orbitals and rank them according to their type (HOMO, LUMO or SOMO) and contribution to the bond (bonding, anti-bonding or non-bonding).
- E (eV): $2s_{\text{N}} = -20,3$ $2p_{\text{N}} = -14,5$ $2s_{\text{O}} = -28,5$ $2p_{\text{O}} = -13,6$

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Question 5. Thermochemistry of NaPF₆ (10% of total; only 10th and 11th grades) Day 1

5.1	5.2	5.3	5.4	5.5	5.6	Total	Total
4	2	2	2	18	12	40	10

The hexafluorophosphate anion (PF₆⁻) forms salts that can be used as electrolytes in lithium-ion and sodium batteries. The thermal and chemical stability of these salts is a critical factor in energy storage technologies. Sodium hexafluorophosphate can be obtained by reacting phosphorus pentachloride, sodium chloride and hydrofluoric acid.

5.1. Represent the Lewis structure of the anion. Say the hybridization of phosphorus and the FPF angle.

5.2. Pose the equation of the salt formation reaction NaPF₆.

The lattice energy is a measure of the stability of ionic solids and can be calculated from the Kapustinskii equation below. *U* is the lattice energy in kJ/mol, *r* and *z* are the radii in pm, and the ions' charges, respectively. $\sum v$ is the total number of ions in the empirical formula.

$$U \approx -1,202 \cdot 10^5 \cdot \frac{\sum v \cdot |z^+||z^-|}{r^+ + r^-} \cdot \left(1 - \frac{34,5}{r^+ + r^-}\right)$$

5.3. Propose an expression to estimate the radius of the PF₆⁻ ion from the covalent and ionic radii.

5.4. Estimate the lattice energy of NaPF₆ if the thermochemical radii of the ions are 102 and 242 pm.

5.5. Use the thermodynamic data **in the table** (energies in kJ/mol, 298 K) and the Born-Haber cycle to calculate the formation enthalpy of NaPF₆. Consider that each P₄ molecule has six P-P bonds.

Sublimation energy of Na(s), ΔH _{sub} (Na)	+108
Primary ionization energy of Na(g), I ₁ (Na)	+496
Sublimation energy of P ₄ (s), ΔH _{sub} (P ₄)	+59
Bond dissociation energy of P-P(g), E(P-P)	+205
Bond dissociation energy of F-F(g), E(F-F)	+155
Energía bond energy of P-F in PF ₆ (g), E(P-F)	+490
Electron affinity energy of the radical PF ₆ (g), ΔH _{AE} (PF ₆)	-770
Lattice energy of NaPF ₆ (s), U(NaPF ₆)	-600
Lattice energy of NaF(s), U(NaF)	-910
Primary electron affinity energy of F(g), A ₁ (F)	-328

5.6. Calculate the enthalpy for the thermic decomposition of the complex salt.



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Question 6. Stoichiometry or chemical calculus (10% of total; only **10th** grade) **Day 1**

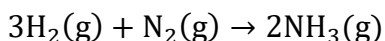
6.1	6.2	6.3	6.4	6.5	Total	Total
12	10	2	12	4	40	10

Agriculture demands large quantities of fertilizer to compensate for the nutrient deficit in soil resulting from year-after-year exploitation. The essential nutrients required in greater quantities include the elements N, P and K. Nitrogen is mainly supplied in the form of the organic compound urea, $\text{CO}(\text{NH}_2)_2$ and the inorganic salts NH_4NO_3 and $(\text{NH}_4)_2\text{HPO}_4$. In turn, these fertilizers are obtained from ammonia, an essential product for modern agriculture.

6.1. Represent the Lewis structure of urea. Propose its electronic distribution by the Valence Bond Theory; consider only valence electrons and hybridize C and N atoms.

6.2. Calculate the nitrogen content (in % by mass) in the three fertilizers and say which one is best suited to supply the most nitrogen.

Ammonia is obtained by reacting dinitrogen and dihydrogen under very high temperatures and pressures ($T \sim 720 \text{ K}$, $P \sim 2000 \text{ atm}$) using a heterogeneous iron catalyst.



Typically, it is marketed as a saturated aqueous solution with an ammonia mass content of 30% and a density of 0.90 g/mL .

6.3. Why does ammonia production require high temperatures and the use of a catalyst?

6.4. Suppose you want to get one ton of ammonia from air (21% O_2 , 79% N_2 , v/v) and hydrogen from electrolysis (decomposition of water by electricity). Calculate the minimum air and water masses required for the process.

6.5. Calculate the volume (in L) of an industrial saturated solution containing one ton of ammonia.

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Day 1. Grade 10th. Questions 1, 4, 5 and 6.